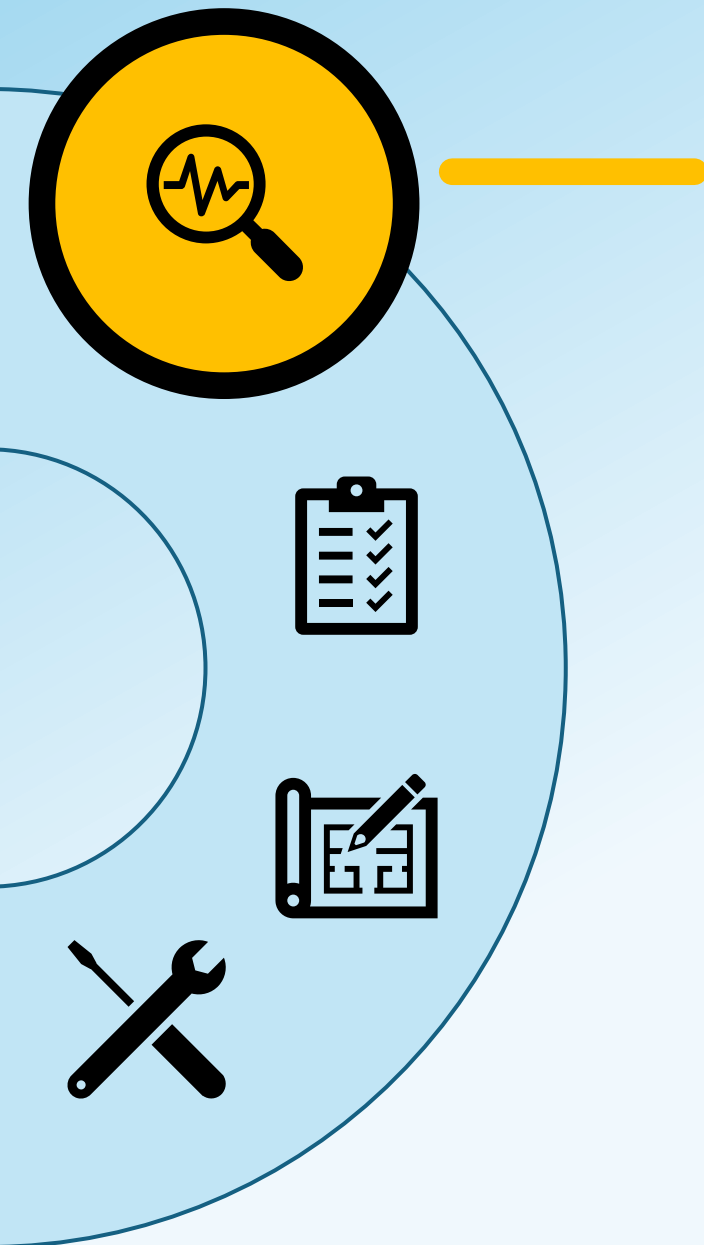


**DESIGN AND IMPLEMENTATION
REAL-TIME LAN-BASED
QUEUING MANAGEMENT &
STATISTICAL REPORTING
SYSTEM
FOR THE BPL-LIEO OF LGU ROSARIO, LA UNION**

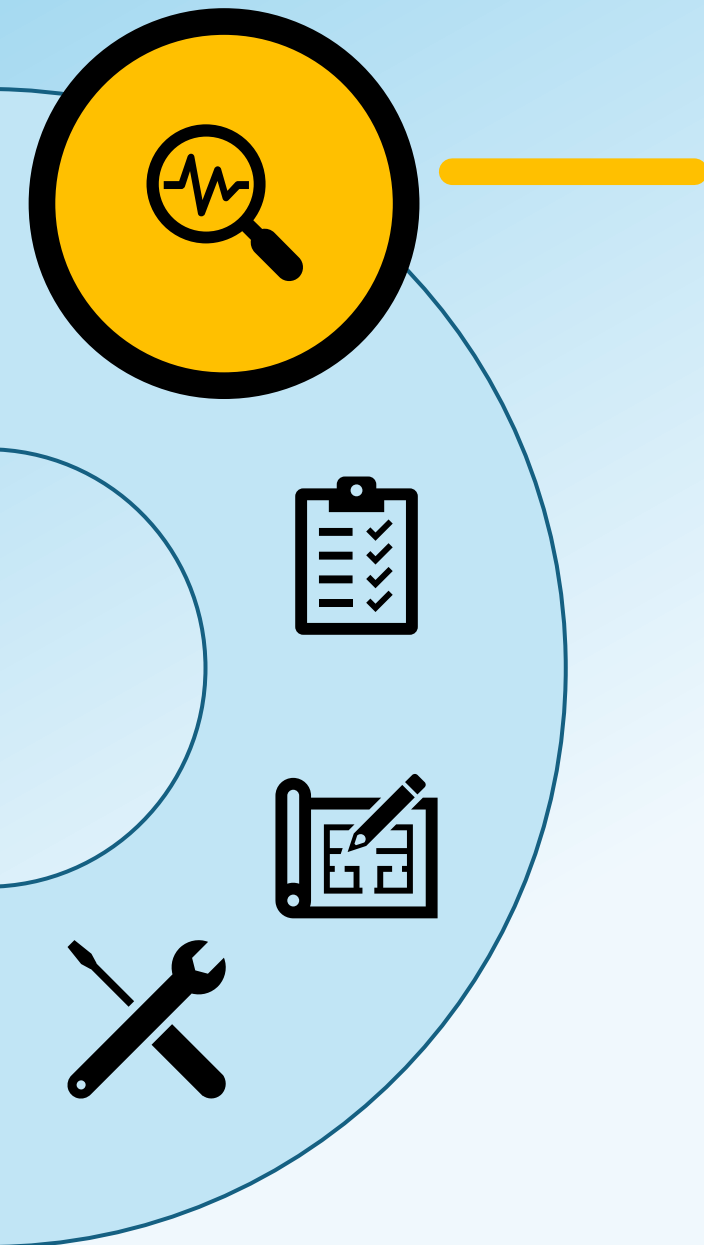
Tandas, Rojean F.
Barbosa, Manrijan M.
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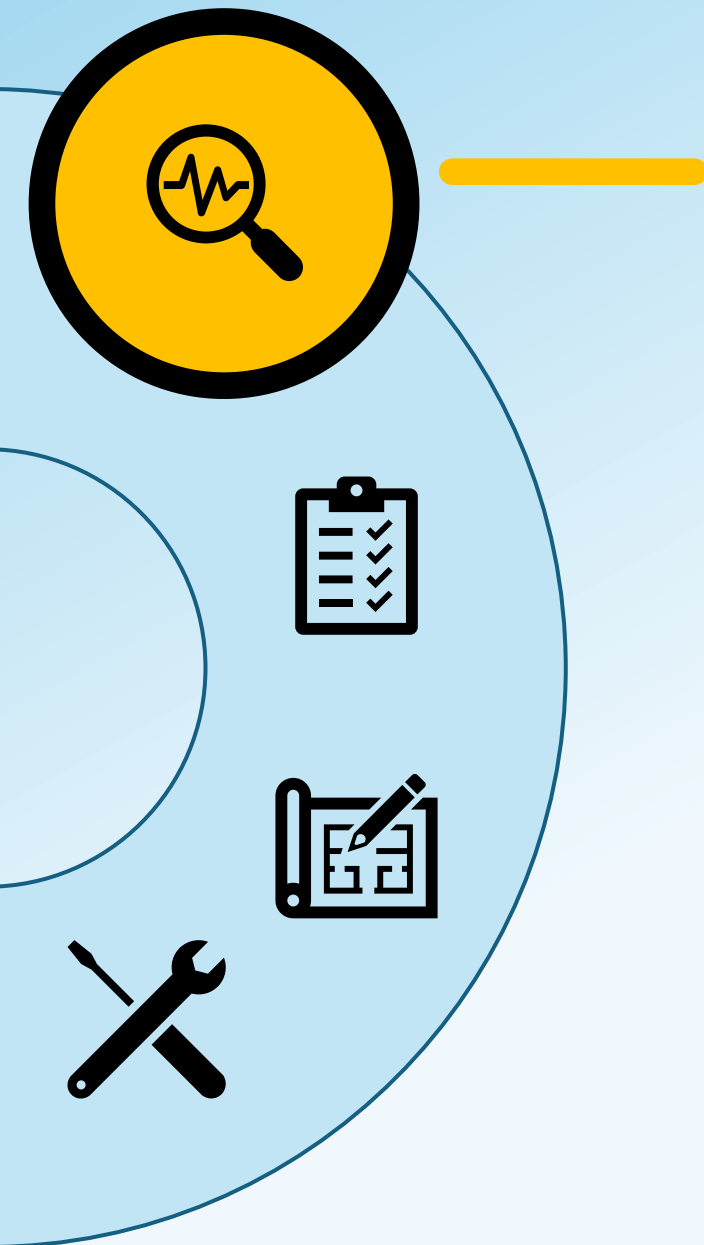
Situation Analysis: The Current State

- **The Context:** BPL-LIEOs accommodate a massive influx of clients during peak business registration and renewal periods.
- **The Current Method:** Heavy reliance on manual queuing and verbal calling.
- **The Bottlenecks:**
 - Disorganized client flow and overcrowding.
 - Client uncertainty regarding queue position.
 - Staff difficulty in simultaneously processing documents, managing the line, and tracking skipped or returning clients.



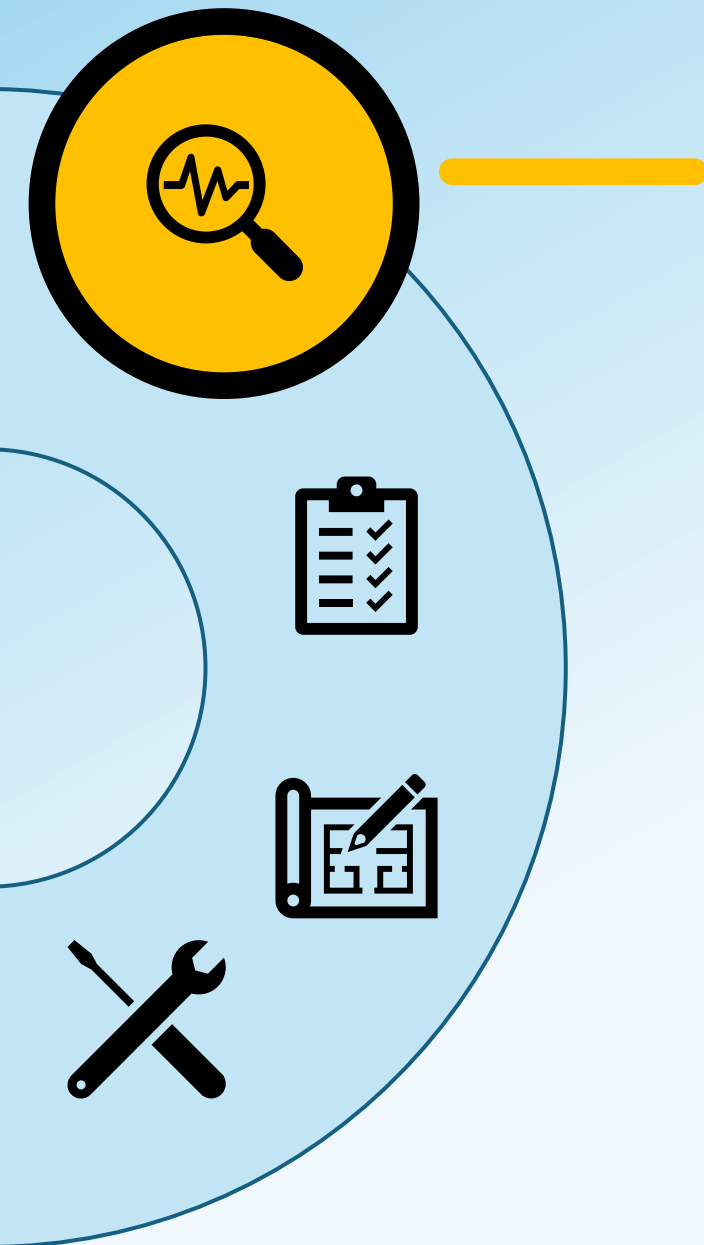
Situation Analysis: The Accessibility Gap

- **The Reality of the Market:** Advanced queuing features (AI forecasting, audit trails) exist in enterprise platforms like Qminder and Wavetec, but they require expensive recurring subscriptions and fast, reliable cloud internet.
- **The LGU Limitation:** Free cloud-based queuing systems exist, but they strip out essential analytics and fail entirely when the internet goes down. Municipal LGUs like Rosario cannot afford enterprise fees nor rely on unstable internet.
- **The True Gap:** There is no free, fully offline, self-hosted queuing system that provides enterprise-grade accountability and analytics tailored for the specific workflow of a local government office.



Situation Analysis: Our Competitive Edge

- **Immutable Audit Trails (Free vs. Paid):** While commercial systems lock staff accountability logs behind expensive "Service Intelligence" paywalls, our system natively will include tamper-evident logging to ensure LGU transparency—at zero cost.
- **Prescriptive Bottleneck Alerts:** Instead of just showing past data, our system will use real-time rolling averages to detect if a specific queue (e.g., Renewals) is moving too slowly, instantly alerting the Admin to open a new window.
- **100% Infrastructure Independence:**
- **Native "Return Tomorrow" Handling:**



Situation Analysis: Our Competitive Edge

- **Immutable Audit Trails (Free vs. Paid):**
- **Prescriptive Bottleneck Alerts:**
- **100% Infrastructure Independence:** By running entirely on a Local Area Network (LAN) utilizing React, Node, and Prisma, we guarantee zero downtime even during municipal internet outages.
- **Native "Return Tomorrow" Handling:** Instead of relying on paid SMS appointment add-ons, we will build priority re-entry directly into the local workflow to accommodate clients with incomplete requirements.



Statement of Objectives

- **General Objective:** To design and implement a Real-Time LAN-Based Queuing Management and Statistical Reporting System specifically adapted for the Rosario BPL-LIEO.
- **Specific Objectives:**
 1. To design and develop an automated LAN-based system.
 2. To test and improve the system's performance, real-time data synchronization, and overall functionality through iterative alpha and beta testing in a local area network environment.
 3. To assess the quality of the developed system in terms of functionality, reliability, usability, efficiency, maintainability, portability, compatibility and security based on the ISO 25010 Software Quality Standards, as assessed by the BPL-LIEO staff and clients.



Research Design

- **Developmental Research:** Systematically designing, building, and refining the queuing software to directly solve the operational bottlenecks of the BPL-LIEO.
- **Descriptive Research:** Utilized to objectively evaluate the system's quality and gather quantitative feedback from both staff and citizens compared to their experience with the manual method.

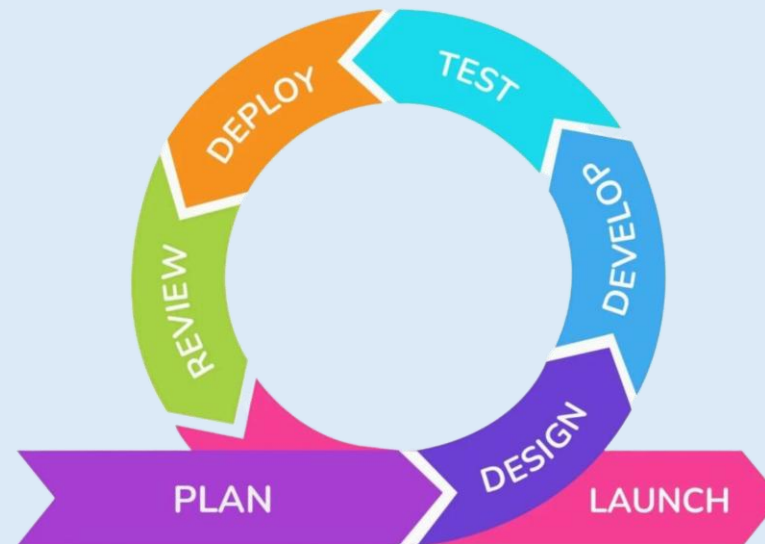


Materials and Procedure: Infrastructure & Tech Stack

- **Network Infrastructure (Star Topology):** A central wireless router connects all local devices (Server, Staff Laptops, TV Display). This ensures that if one staff laptop disconnects, the rest of the system remains fully operational.
- **Proposed Fault Tolerance (100% Uptime):**
 - **PM2 Process Manager:** Automatically restarts the Node.js server within a second if a crash occurs.
 - **SQLite WAL Mode:** "Write-Ahead Logging" prevents database corruption during sudden power outages.
 - **Socket.io Auto-Reconnect:** Client screens seamlessly re-sync without requiring manual page refreshes if the network drops.
- **Software Stack:** React.js (Frontend), Node.js & Express.js (Backend), Prisma ORM (Database).

Materials and Procedure: SDLC

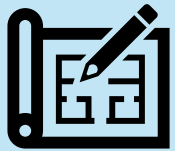
- **Approach:** Agile Software Development Life Cycle (SDLC).
- **Why Agile?** It allows us to build the system incrementally in sprints, ensuring we can easily adapt the software based on direct feedback from BPL-LIEO personnel during development.
- **Phases:** Plan, Design, Develop, Test, Deploy and Review.



Materials and Procedure: System Design

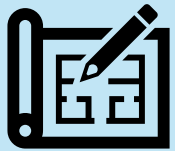
➤ Visualizing the Solution:

1. BPMN Diagram: Maps the step-by-step physical and digital interactions from the citizen's arrival to departure.
2. Entity-Relationship Diagrams (ERD): Ensuring robust transaction data storage.
3. Data Flow & Use-Case Diagrams: Defining clear roles for Citizens vs. Staff.



Materials and Procedure: Testing Procedures

- **Continuous Testing:** Ensuring code quality and real-time Socket.io stability.
- **Alpha Testing:** Conducted in a controlled lab environment by developers to iron out initial logic and UI bugs.
- **Beta Testing:** The crucial phase, deployed on the actual BPL-LIEO LAN environment and tested by actual personnel to validate network performance and usability under real-world conditions.





Data Gathered

- **Assessment Tool:** A customized survey questionnaire based on the ISO/IEC 25010 Software Quality Model (adapted from Malaya et al., 2022).
- **Respondents (Total N = 60):**
 - **BPL-LIEO Staff:** 10 respondents (Total Enumeration) – evaluating operational efficiency and staff controls.
 - **Citizens / Clients:** 50 respondents (Convenience Sampling) – evaluating public monitor visibility and process transparency.



Data Analysis

- **Statistical Tool:** Weighted Mean
- **Evaluation Scale:** 5-point Likert Scale assessing seven ISO/IEC 25010 characteristics (Functionality, Usability, Efficiency, Reliability, Security, Maintainability, Portability).
- **Goal:** To determine the descriptive equivalent rating (from Poor to Excellent) and definitively prove the system's acceptability and superiority over the manual method.

THANK YOU!

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